BOUNDED ABOVE ACYCLIC SUBCOMPLEXES OF BOUNDED SIZE INSIDE ANY ACYCLIC COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete construction of both assertions, with the closing-up of the chosen surjections into a subcomplex carried out in the span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- A generator of the category. Same chapter, the existence of a generator of a Grothendieck abelian category. Supplies the object U used to define the cardinal bound.
- The surjection of bounded size. Same chapter, Lemma lemma-surjection-bounded-size. Supplies the function c and the statement that every object receives a surjection from a coproduct of at most c many copies of the generator, which the author iterates along the complex.

Source reference. Original lemma 079J: sources/stacks-injectives/pinned-original-source.tex, source lines 1787–1787. Navigation pointer invoked at proof line 1824; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ be a Grothendieck abelian category. There exists a cardinal κ such that given any acyclic complex $M^\bullet$ we have*

(1) *if $M^\bullet$ is nonzero, there is a nonzero subcomplex $N^\bullet$ which is bounded above, acyclic, and $|N^n| \leq \kappa$,*

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(2) *there exists a surjection of complexes*

$$\bigoplus\nolimits_{i\in I} M_i^\bullet \longrightarrow M^\bullet$$

where $M_i^\bullet$ is bounded above, acyclic, and $|M_i^n| \leq \kappa$.

Proof. Choose a generator U of $\mathcal{A}$. Denote c the function of Lemma 079J. Set $\kappa = \sup\{c^n(|U|), n = 1, 2, 3, \ldots\}$. Let $n \in \mathbf{Z}$ and let $\psi : U \to M^n$ be a morphism. In order to prove (1) and (2) it suffices to prove there exists a subcomplex $N^\bullet \subset M^\bullet$ which is bounded above, acyclic, and $|N^m| \leq \kappa$, such that ψ factors through N^n. To do this set $N^n = \mathrm{Im}(\psi)$, $N^{n+1} = \mathrm{Im}(U \to M^n \to M^{n+1})$, and $N^m = 0$ for $m \geq n + 2$. Suppose for some $k \leq n$ we have constructed $N^m \subset M^m$ for all $n > m \geq k$ such that

(1) $\mathrm{d}(N^m) \subset N^{m+1}$ for all $m \geq k$,
(2) $\mathrm{Im}(N^{m-1} \to N^m) = \mathrm{Ker}(N^m \to N^{m+1})$ for all $m \geq k + 1$, and
(3) $|N^m| \leq c^{\max\{n-m,0\}}(|U|)$ for all $m \geq k$.

Because $M^\bullet$ is acyclic, we see that the subobject $\mathrm{d}^{-1}(\mathrm{Ker}(N^k \to N^{k+1})) \subset M^{k-1}$ surjects onto $\mathrm{Ker}(N^k \to N^{k+1})$. Thus we can choose $N^{k-1} \subset M^{k-1}$ surjecting onto $\mathrm{Ker}(N^k \to N^{k+1})$ with $|N^{k-1}| \leq c^{n-k+1}(|U|)$ by Lemma 079J. The proof is finished by induction on k. □